Chapter 5
Differentiation

Note 3

Lecturer: Dr. Rajitha Ranasinghe Academic Year: 2024/2025

5.3 : The Mean Value Theorem

The Mean Value Theorem makes a precise statement about the relationship between the derivative of a function and the slope of one of its chords. It was discovered by the following geometric reasoning. Suppose that f is differentiable on (a, b). Since the graph of f on (a, b) has a tangent at each of its points, it seems likely that the slope of the chord through the points $(a, f(a))$ and $(b, f(b))$ equals the slope $f'(c)$ for some value of $c \in (a, b)$.

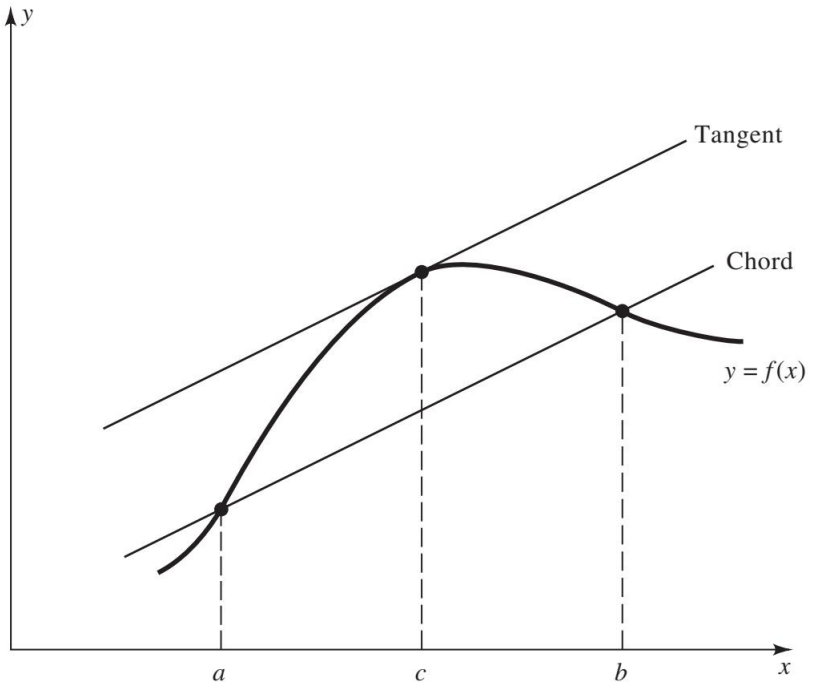


Figure 5.3.1

We begin with a special case.

Lemma : Rolle's Theorem

Suppose that $a, b \in \mathbb{R}$ with $a < b$. If f is continuous on $[a, b]$, differentiable on (a, b), and if $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Proof.

- By the **Extreme Value Theorem**, f has a finite maximum M and a finite minimum m attains on $[a, b]$.
- If $M = m$, then f is constant on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$.
- Suppose that $M \neq m$. Since $f(a) = f(b)$, f must assume one of the values M or m at some point $c \in (a, b)$.
- By symmetry, we may suppose that $f(c) = M$. (That is, if we can prove the theorem when $f(c) = M$, then a similar proof establishes the theorem when $f(c) = m$.)
- Since M is the maximum of f on $[a, b]$, we have
$$f(c + h) - f(c) \leq 0$$
for all h which satisfy $c + h \in (a, b)$.
- In the case $h > 0$ this implies
$$f'(c) = \lim_{h \to 0^+} \frac{f(c + h) - f(c)}{h} \leq 0 \ ,$$
and in the case $h < 0$ this implies
$$f'(c) = \lim_{h \to 0^-} \frac{f(c + h) - f(c)}{h} \geq 0 \ .$$
- It follows that $f'(c) = 0$.

□

Note : The proof of the Rolle's Theorem proves a well-known result: The extreme values of a differentiable function on an open interval occur at critical points (i.e., at points where f' is zero).

Remark 1 : The *continuity* hypothesis in the Rolle's Theorem cannot be dropped at even one point in $[a, b]$.

Proof. The function $f(x) = \begin{cases} x & x \in [0, 1), \\ 0 & x = 1, \end{cases}$ is continuous on $[0, 1)$, differentiable on $(0, 1)$, and $f(0) = f(1) = 0$, but $f'(x)$ is never zero. □

Remark 2 : The *differentiability* hypothesis in Rolle's Theorem cannot be dropped even at one point in (a, b).

Proof. The function $f(x) = |x|$ is continuous on $[-1, 1]$, differentiable on $(-1, 1)\backslash\{0\}$, and $f(-1) = f(1)$, but $f'(x)$ is never zero. □

We shall use Rolle's Theorem to obtain several useful results. The first is a pair of "Mean Value Theorems".

Theorem 5.3.1 : Suppose that $a, b \in \mathbb{R}$ with $a < b$.

(i). Generalized Mean Value Theorem : If f, g are continuous on $[a, b]$ and differentiable on (a, b), then there is a $c \in (a, b)$ such that $\boxed{g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))}$

(ii). Mean Value Theorem : If f is continuous on $[a, b]$ and differentiable on (a, b), then there is a $c \in (a, b)$ such that $\boxed{f(b) - f(a) = f'(c)(b - a)}$

Proof.

(i).
- Set $h(x) = f(x)(g(b) - g(a)) - g(x)(f(b) - f(a))$ and note that $h'(x) = f'(x)(g(b) - g(a)) - g'(x)(f(b) - f(a))$.
- So h is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = h(b)$.
- Thus by the Rolle's theorem, $h'(c) = 0$ for some $c \in (a, b)$, which gives the desired result.

(ii). Set $g(x) = x$ and apply part **(i)**. (For a geometric interpretation of this result, see the opening paragraph of this section and **Figure 6.3.1**). □

Note :

- The **Generalized Mean Value Theorem** is also called ***Cauchy's Mean Value Theorem***. It is usually essential when comparing derivatives of two functions simultaneously, using higher-order derivatives to approximate functions, and studying certain kinds of generalized derivatives.
- The **Mean Value Theorem** is most often used to extract information about f from f'. Perhaps the best-known result of this type is the criterion for deciding when a differentiable function increases. To prove this result, we begin with the following definition.

Definition 5.3.1 :

Let E be a non-empty subset of $\mathbb{R}$ and $f : E \to \mathbb{R}$.

(i). f is said to be *increasing* (respectively, *strictly increasing*) on E if and only if $x_1, x_2 \in E$ and $x_1 < x_2$ imply $f(x_1) \leq f(x_2)$ [respectively, $f(x_1) < f(x_2)$].

(ii). f is said to be *decreasing* (respectively, *strictly decreasing*) on E if and only if $x_1, x_2 \in E$ and $x_1 < x_2$ imply $f(x_1) \geq f(x_2)$. [respectively, $f(x_1) > f(x_2)$].

(iii). f is said to be *monotone* (respectively, *strictly monotone*) on E if and only if f is either decreasing or increasing (respectively, either strictly decreasing or strictly increasing) on E.

Thus, although $f(x) = x^2$ is strictly monotone on $[0, 1]$, and on $[-1, 0]$, it is not monotone on $[-1, 1]$. Monotone functions are important from both a theoretical and a practical point of view. Thus it will come as no surprise that the next result is very important and widely used.

Theorem 5.3.2 : Suppose that $a, b \in \mathbb{R}$, with $a < b$, that f is continuous on $[a, b]$ and that f is differentiable on (a, b).

(i). If $f'(x) > 0$ [respectively, $f'(x) < 0$] for all $x \in (a, b)$, then f is strictly increasing (respectively, strictly decreasing) on $[a, b]$.

(ii). If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.

(iii). If g is continuous on $[a, b]$ and differentiable on (a, b), and if $f'(x) = g'(x)$ for all $x \in (a, b)$, then $f - g$ is constant on $[a, b]$.

Proof.

(i). Let $a \leq x_1 < x_2 \leq b$. By the Mean Value Theorem, there is a $c \in (a, b)$ such that $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. Thus, $f(x_2) > f(x_1)$ when $f'(c) > 0$ and $f(x_2) < f(x_1)$ when $f'(c) < 0$.

(ii). Note that if $f' = 0$, then by the proof of part i), f is both increasing and decreasing, and hence constant on $[a, b]$.

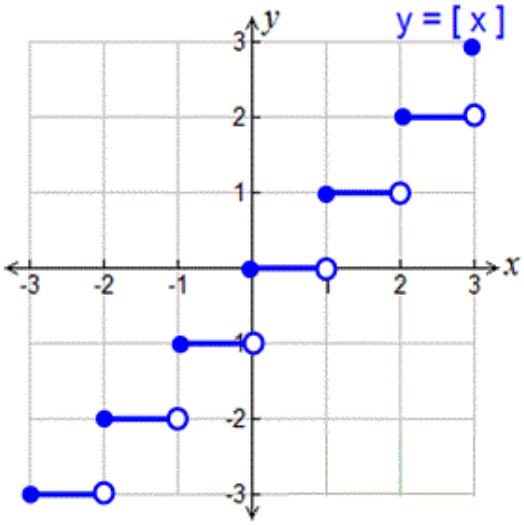
(iii). Apply **(ii)** to the function $f - g$. □

Remark :

- **Theorem 5.3.2** is significant because it makes checking whether a differentiable function is monotone straightforward. However, there are many non-differentiable functions which are monotone. For example, the greatest integer function

$$f(x) = *x := n, \quad n \leq x < n+1, \quad n \in \mathbb{Z},$$

is increasing on $\mathbb{R}$ but neither continuous nor differentiable.



- How do non-differentiable monotone functions behave ? The following result shows that, similar to the greatest integer function, any function that is monotone on an interval always has left and right limits. This is a function analogue of the **Monotone Convergence Theorem**.

Theorem 5.3.3 : Suppose that f is increasing on $[a, b]$.

(i). If $c \in [a, b)$, then $f(c^+)$ exists and $f(c) \leq f(c^+)$.

(ii). If $c \in (a, b]$, then $f(c^-)$ exists and $f(c^-) \leq f(c)$.

Proof.

- By symmetry it suffices to show that $f(c^-)$ exists and satisfies $f(c^-) \leq f(c)$ for any fixed $c \in (a, b]$.

- Set $E = f((a, c))$ and $s = \sup E$. Since f is increasing, $f(c)$ is an upper bound of E.
- Hence, s is a finite real number which satisfies $s \leq f(c)$.
- Given $\varepsilon > 0$, choose by the **Approximation Property** an $x_0 \in (a, c)$ such that $s - \varepsilon < f(x_0) \leq s$. Since f is increasing,
$$s - \varepsilon < f(x_0) \leq f(x) \leq s \quad \text{for all} \quad x_0 < x < c.$$
- Therefore, $f(c^-)$ exists and satisfies $f(c^-) = s \leq f(c)$.

□

Theorem 5.3.2 (i) can be used for tasks other than finding intervals on which a given function is increasing. The following example shows how to use it to compare one function with another.

Example 1 Prove that $1 + x < e^x$ for all $x > 0$.

Proof. Let $f(x) = e^x - x$, and observe that $f'(x) = e^x - 1 > 0$ for all $x > 0$ It follows from **Theorem 5.3.2 (i)** that $f(x)$ is strictly increasing on $(0, \infty)$. Thus $e^x - x = f(x) > f(0) = 1$ for $x > 0$. In particular, $e^x > x + 1$ for $x > 0$. □

We close this section with some optional results which further explore the mean value concept. Our first result shows how $(1 + x)^\alpha$ is related to $1 + \alpha x$.

Theorem 5.3.4 : Bernoulli's Inequality

Let α be a positive real number. If $0 < \alpha \leq 1$, then $(1 + x)^\alpha \leq 1 + \alpha x$ for all $x \in [-1, \infty)$, and if $\alpha \geq 1$, then $(1 + x)^\alpha \geq 1 + \alpha x$ for all $x \in [-1, \infty)$.

Proof.

- The proofs of these inequalities are similar. We present the details only for the case $0 < \alpha \leq 1$.
- Fix $x \geq -1$ and let $f(t) = t^\alpha, t \in [0, \infty)$.
- Since $f'(t) = \alpha t^{\alpha-1}$, it follows from the **Mean Value Theorem** (applied to $a = 1$ and $b = 1+x$) that
$$f(1 + x) - f(1) = \alpha x c^{\alpha-1} \tag{12}$$
for some c between 1 and $1 + x$.

- **Case 1 :** $\boxed{x > 0}$ Then $c > 1$. Since $0 < \alpha \leq 1$ implies $\alpha - 1 \leq 0$, it follows that $c^{\alpha-1} \leq 1$, hence $xc^{\alpha-1} \leq x$. Therefore, we have by (12) that

$$(1+x)^\alpha = f(1+x) = f(1) + \alpha x c^{\alpha-1} \leq f(1) + \alpha x = 1 + \alpha x \tag{13}$$

as required.

- **Case 2 :** $\boxed{-1 \leq x \leq 0}$ Then $c \leq 1$ so $c^{\alpha-1} \geq 1$. But since $x \leq 0$, it follows that $xc^{\alpha-1} \leq x$ as before and we can repeat (13) to obtain the same conclusion.

□

We will now use Bernoulli's Inequality to show once again that vague reasoning can produce wrong conclusions. To see why, assuming that $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$ exists, what do you think its limit is? Vague reasoning that well over half your class would agree with: Since $1 + 1/n$ gets near 1 and $1^\alpha = 1$ for all $\alpha \in \mathbb{R}$, the limit should be 1, right ? Absolutely not !

Example 2 Prove that the sequence $(1+1/n)^n$ is increasing, as $n \to \infty$, and its limit L satisfies $2 < L \leq 3$. (The limit L turns out to be an irrational number, the natural base $e = 2.718281828459\cdots$.)

Proof. The sequence $(1 + 1/n)^n$ is increasing, since by **Bernoulli's Inequality**,

$$\left(1 + \frac{1}{n}\right)^{n/(n+1)} \leq \left(1 + \frac{1}{n+1}\right)$$

To prove that this sequence is bounded above, observe by the Binomial Formula that

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k} \left(\frac{1}{n}\right)^k.$$

Now,

$$\binom{n}{k} \left(\frac{1}{n}\right)^k = \frac{n(n-1)\ldots(n-k+1)}{n^k} \cdot \frac{1}{k!} \leq \frac{1}{k!} \leq \frac{1}{2^{k-1}}$$

for all $k \in \mathbb{Z}^+$. It follows that (Use Mathematical Induction)

$$2 = \left(1 + \frac{1}{1}\right) < \left(1 + \frac{1}{n}\right)^n \leq 1 + 1 + \sum_{k=1}^{n-1} \frac{1}{2^k} = 3 - \frac{1}{2^{n-1}} < 3$$

for $n > 1$. Hence, by the **Monotone Convergence Theorem**, the limit L exists and satisfies $2 < L \leq 3$

□

The last result in this section shows that although a differentiable function might not be continuously differentiable, its derivative does satisfy an intermediate value theorem. (This result is sometimes called **Darboux's Theorem**.)

Theorem 5.3.5 : Intermediate Value Theorem for Derivatives

Suppose that f is differentiable on $[a, b]$ with $f'(a) \neq f'(b)$. If $y_0 \in \mathbb{R}$ which lies between $f'(a)$ and $f'(b)$, then there is an $x_0 \in (a, b)$ such that $f'(x_0) = y_0$.

Proof.

- Suppose that y_0 lies between $f'(a)$ and $f'(b)$.
- By symmetry, we may suppose that $f'(a) < y_0 < f'(b)$.
- Set $F(x) = f(x) - y_0 x$ for $x \in [a, b]$ and observe that F is differentiable on $[a, b]$.
- Hence, by the **Extreme Value Theorem**, F has an absolute minimum, say $F(x_0)$, on $[a, b]$.
- Now $F'(a) = f'(a) - y_0 < 0$, so $F(a + h) - F(a) < 0$ for $h > 0$ sufficiently small.
- Hence $F(a)$ is NOT the absolute minimum of F on $[a, b]$. Similarly, $F(b)$ is not the absolute minimum of F on $[a, b]$. Hence, the absolute minimum $F(x_0)$ must occur on (a, b); that is, $x_0 \in (a, b)$ and $F'(x_0) = 0$.

□

Exercises

(1). Suppose that $f, g : [a, b] \to \mathbb{R}$. Decide which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

(a). If f and g are increasing on $[a, b]$, then $f + g$ is increasing on $[a, b]$.

Solution : TRUE

- Let $x, y \in [a, b]$ be such that $x < y$.
- Since f and g are increasing on $[a, b]$, we have $f(x) \leq f(y)$ and $g(x) \leq g(y)$.
- Adding the two inequalities above yield $f(x) + g(x) \leq f(y) + g(y)$. i.e., $f + g$ is increasing on $[a, b]$.

(b). If f and g are increasing on $[a, b]$, then fg is increasing on $[a, b]$.

Solution : FALSE

Counter Example : Note that $f(x) = g(x) = x$ are increasing on $[-1, 0]$ but $f(x) \cdot g(x) = x^2$ is decreasing on $[-1, 0]$.

(c). If f is differentiable on (a, b) and $\lim_{x \to a^+} f(x)$ exists and is finite, then for each $x \in (a, b)$ there is a c between a and x such that $f(x) - f(a^+) = f'(c)(x - a)$.

Solution : TRUE

- Define a function g as follows:

$$g(x) := \begin{cases} f(x) & \text{for } x \in (a, b), \\ f(a^+) & \text{for } x = a. \end{cases}$$

- Note that g is continuous on $[a, x]$ for every $x \in (a, b)$.
- By the Mean Value THeorem, $\exists c \in (a, b)$ such that

$$f(x) - f(a^+) = g(x) - g(a) = g'(c)(x - a) = g'(c)(x - a).$$

(d). If f and g are differentiable on $[a,b]$ and $|f'(x)| \leq 1 \leq |g'(x)|$ for all $x \in (a,b)$, then $|f(x)-f(a)| \leq |g(x)-g(a)|$ for all $x \in [a,b]$.

Solution : TRUE

For any $x \in (a,b)$, by the Mean value theorem, $\exists c, d \in (a,x)$ such that

$$|f(x)-f(a)| = (x-a)|f'(c)| \leq (x-a)|g'(d)| = |f(x)-f(a)|.$$

(2). Prove that each of the following inequalities holds.

(a). $2x + 0.7 < e^x$ for all $x \geq 1$.

Proof :

blue Define $f(x) = e^x - 2x - 0.7$.

blue For $x \geq 1$, $f'(x) = e^x - 2 > 0$; hence f is strictly increasing on $[1,\infty)$ by monotonicity.

blue Therefore $f(x) \geq f(1) = e - 2.7 > 0$ for all $x \geq 1$.

□

(b). $\log x < \sqrt{x} - 0.6$ for all $x \geq 4$.

Proof :

blue Define $f(x) = \sqrt{x} - \log x - 0.6$ on $[4,\infty)$.

blue Compute $f'(x) = \dfrac{1}{2\sqrt{x}} - \dfrac{1}{x} = \dfrac{x - 2\sqrt{x}}{2x\sqrt{x}} \geq 0$ for $x \geq 4$.

blue Hence f is increasing on $[4,\infty)$ and $f(x) \geq f(4) = 2 - \log 4 > 0$. So inequality holds.

□

(c). $\sin^2 x \leq 2|x|$ for all $x \in \mathbb{R}$.

Proof :

blue Define $f(x) = 2|x| - \sin^2 x$.

blue For $x \geq 0$: $f(x) = 2x - \sin^2 x$ with $f'(x) = 2 - 2\sin x \cos x \geq 0$ since $|\sin(2x)| \leq 1$.

Thus $f(x) \geq 0$.

blue For $x < 0$: $\sin^2 x = \sin^2(-x) \leq 2|-x| = 2|x|$. Hence the inequality holds.

□

(d). $1 - \sin x \leq e^x$ for all $x \geq 0$.

Proof :

blue Define $f(x) = e^x - 1 + \sin x$.

blue $f'(x) = e^x + \cos x \geq 1 + \cos x \geq 0$ for all $x \geq 0$.

blue Thus f is increasing on $[0, \infty)$ and $f(x) \geq f(0) = 0$.

□

(3). Suppose that $I = (0, 2)$, that f is continuous at $x = 0$ and $x = 2$, and that f is differentiable on I. If $f(0) = 1$ and $f(2) = 3$, prove that $1 \in f'(I)$.

Proof : By the Mean Value Theorem, $2 = f(2) - f(0) = 2f'(c)$ for some $c \in (0, 2)$. Thus $f'(c) = 1$, i.e., $1 \in f'(0, 2)$.

□

(4). Note that $r \in \mathbb{R}$ is called a *root* of a function f if and only if $f(r) = 0$. Show that if f is differentiable on $\mathbb{R}$, then its derivative f' has at least one root between any two roots of f.

Proof : If a and b $(a \neq b)$ are roots of f, by the Mean Value Theorem it follows that, $0 = f(b) - f(a) = (b - a)f'(c)$ for some $c \in (a, b)$, which implies $f'(c) = 0$. Hence c is a root of f'. □

(5). Suppose that f is differentiable on $\mathbb{R}$. If $f(0) = 1$ and $|f'(x)| \leq 1$ for all $x \in \mathbb{R}$, prove that $|f(x)| \leq |x| + 1$ for all $x \in \mathbb{R}$.

Proof :

- Apply MVT on $[0, x]$ (or $[x, 0]$ if $x < 0$): $\exists\ c$ between 0 and x with $f(x) - 1 = f'(c)(x - 0)$.
- Take absolute values: $|f(x) - 1| = |f'(c)||x| \leq |x|$.
- Thus $|f(x)| \leq |f(x) - 1| + |1| \leq |x| + 1$.

□

(6). Suppose that f is differentiable on (a, b), continuous on $[a, b]$, and that $f(a) = f(b) = 0$. Prove that if $f(c) \neq 0$ for some $c \in (a, b)$, then there exist $x_1, x_2 \in (a, b)$ such that $f'(x_1) > 0$ and $f'(x_2) < 0$.

Proof :

- Apply MVT on $[a, c]$: $\exists x_1 \in (a, c)$ with $f(c) = f(c) - f(a) = (c - a)f'(x_1)$.
- Apply MVT on $[c, b]$: $\exists x_2 \in (c, b)$ with $-f(c) = f(b) - f(c) = (b - c)f'(x_2)$.
- Note that $c - a > 0$ and $b - c > 0$
- If $f(c) > 0$: $f'(x_1) > 0$ and $f'(x_2) < 0$.
- If $f(c) < 0$: $f'(x_1) < 0$ and $f'(x_2) > 0$.
- In both cases, $f'(x_1)$ and $f'(x_2)$ have opposite signs.

□

(7). Suppose that f is twice differentiable on (a, b) and that there are points $x_1 < x_2 < x_3$ in (a, b) such that $f(x_1) > f(x_2)$ and $f(x_3) > f(x_2)$. Prove that there is a point $c \in (a, b)$ such that $f''(c) > 0$.

Proof :

- Apply MVT to f on $[x_1, x_2]$: $\exists\ c_1 \in (x_1, x_2)$ with $f'(c_1) = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} < 0$.
- Apply MVT to f on $[x_2, x_3]$: $\exists\ c_2 \in (x_2, x_3)$ with $f'(c_2) = \dfrac{f(x_3) - f(x_2)}{x_3 - x_2} > 0$.
- Apply MVT to f' on $[c_1, c_2]$: $\exists\ c \in (c_1, c_2)$ with $f''(c) = \dfrac{f'(c_2) - f'(c_1)}{c_2 - c_1} > 0$.

□